

Unit 9 Trigonometric Ratios & Functions Study Guide

Objectives...Remember that big colored sheet of paper you taped into your notebook at the beginning of the unit??

A. Trigonometric Ratios

- ☐ I can evaluate trigonometric ratios of acute angles.
- ☐ I can find unknown side lengths and angles using trigonometry.
- ☐ I can use trigonometric functions to solve real-life problems.
- ☐ I can draw angles measured in degrees and radians in standard position.
- ☐ I can find and use coterminal angles.
- ☐ I can evaluate trig ratios of any angle.
- ☐ I can find and use reference angles to evaluate trigonometric ratios.
- ☐ I know the unit circle.

B. Graphing Trigonometric Functions

- ☐ I can graph sine and cosine functions.
- ☐ I can graph transformations of sine and cosine functions.
- ☐ I can graph transformations of the tangent function.
- ☐ I can describe the period, amplitude, midline and phase shift of trigonometric graphs.

Topics Covered

- ★ Lesson 1 - Right Triangle Trigonometry
- ★ Lesson 2 - Angles & Radian Measures
- ★ Lesson 3 - Trigonometry Functions of Any Angle
- ★ Lesson 4 - Graphing Sine & Cosine Functions
- ★ Lesson 5 - Modeling Sine & Cosine Functions

C. Trigonometric Identities & Modeling

- ☐ I can write and solve trigonometric equations to solve problems.

Essential Questions

1. What is a trigonometric function?
2. What are the six trigonometric functions?
3. How can I evaluate trigonometric functions?
4. What is the unit circle?
5. How can I use the unit circle?
6. How do we graph trigonometric functions?
7. How do we solve trigonometric equations?

How can I practice/study for this assessment?

Your notes, practice, and homework creates a "practice guide"!

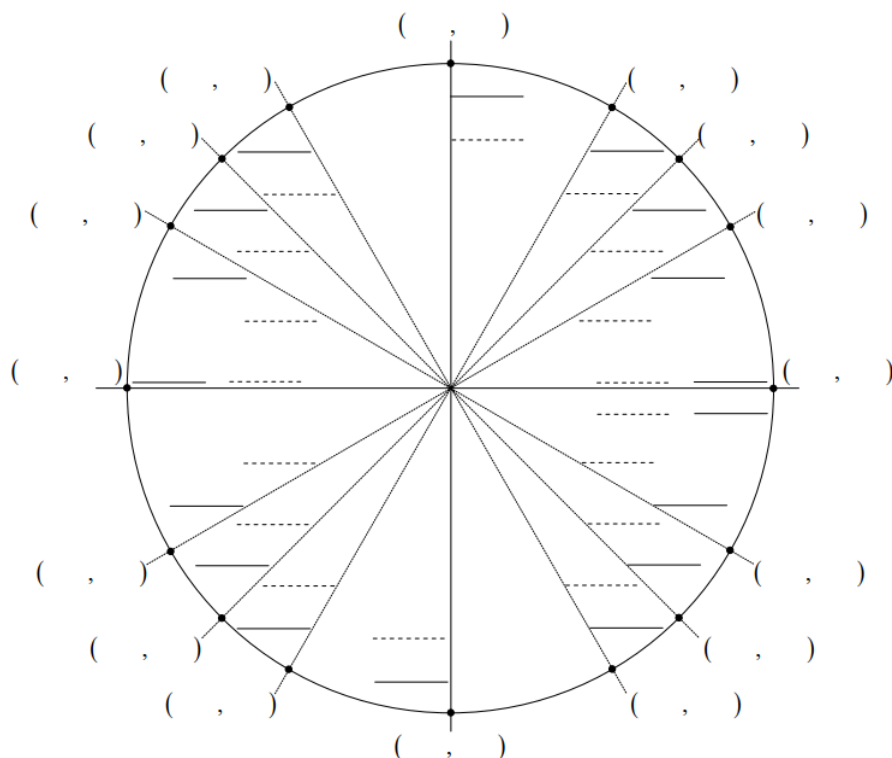
1. Complete any unfinished practice problems
2. Redo practice problems
3. DeltaMath
4. Visit our Google Classroom page for digital versions of in class activities!!
5. Visit our Google Classroom page for any additional digital versions of practice!!

Unit 9 Trigonometric Ratios & Functions Practice Guide

Complete the following problems in your HOMEWORK notebook.

****Challenge Problems.**

- The unit circle is built using _____. The hypotenuse always measures _____, which is also the length of the _____ of the _____ circle.
- On the Unit circle what does (x, y) represent in terms of the trig? How do you calculate $\tan \theta$, since it is not on the circle?
- Without looking at references fill in the Unit Circle for (Degrees, Radians, and Ordered Pairs)



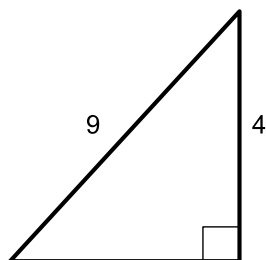
- Given the point $(7, -3)$, find all six trigonometric ratios.
- (skip)**
- What is a reference angle?
- State the reference angles of each of the following.
 - 259°
 - -132°
 - $\frac{5\pi}{7}$
 - $-\frac{3\pi}{5}$
- For the following angle measures:
 - Identify the quadrant it falls within.
 - Draw the angle in standard position (label initial and terminal sides).
 - Calculate the reference angle.
 - Calculate two coterminal angles where one is of opposite sign.
 - Convert the measurement.
- State the quadrant where the following occurs:

a. $\cos \theta > 0$, $\tan \theta < 0$

b. $\sec \theta < 0$, $\sin \theta > 0$

c. $\cot \theta < 0$, $\csc \theta < 0$

10. Solve the triangle. Give exact and rounded (to the nearest hundredth) answers.



11. Use your calculator to evaluate (to the nearest thousandth) the following:

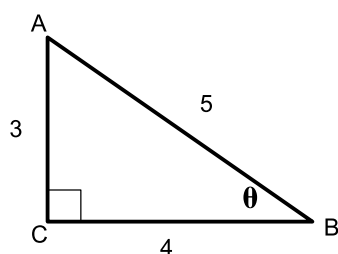
a. $\sin(47^\circ)$

b. $\cos(0.34)$

c. $\sin(0.5)$

12. If $\cos \theta = \frac{3}{7}$ and θ is in Quadrant IV, find the other five trigonometric ratios in exact form.

13. **True or false.** If the statement is false, correct it so that it is true or explain why it is false.



a. $\sin \theta = \frac{3}{5}$

b. $\triangle ABC$ is a 30-60-90 triangle.

c. $\triangle ABC$ is a 45-45-90 triangle.

d. $\cos B = \sin A$

e. $\sin B = \sin A$ since $m\angle A + m\angle B = 180^\circ$

14. **Given $\sin Q = 0.1126$ and $\cos P = 0.1126$, then what is known about $\angle Q$ and $\angle P$?

15. Use DeltaMath for practice evaluating trig ratios of angles on the unit circle.

16. Solve for θ on the unit circle. Provide both radians and degrees for θ .

Example: $\sin \theta = -\frac{\sqrt{2}}{2}$ and $0 < \theta \leq 360^\circ$

17. Given $f(x) = -2 \sin(3(x + \frac{\pi}{6})) + 1$, identify the period, maximum, minimum, amplitude, midline, domain, and range. Then, sketch a graph of the function (including at least one period).

18. Given $f(x) = \frac{1}{2} \cos(\frac{1}{4}(x - 2\pi))$, identify the period, maximum, minimum, amplitude, midline, domain, and range. Then, sketch a graph of the function (including at least one period).

19. In a predator/prey model, the predator population is modeled by the function: $p = 900\cos(\frac{2\pi}{3}t) + 8000$ where t is measured in years.

a. Find the length of time between successive periods of maximum population.

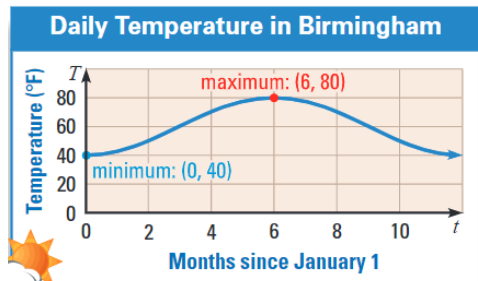
b. When does the population first reach 5000?

c. What is the minimum population? When does this occur in the first cycle?

d. What is the population after 100 years?

20. A buoy bobs up and down as waves go past. The vertical displacement y (in feet) of the buoy with respect to sea level can be modeled by $y = 1.75 \cos \frac{\pi}{3}t$, where t is the time (in seconds). Find and interpret the period and amplitude in context of the problem. Then, graph the function.
21. Two people swing jump ropes. The highest point in the middle of each rope is 80 inches above the ground, and the lowest point is 2 inches. The rope makes 2 revolutions per second. Write a model for the height h (in inches) of a rope as a function of time t (in seconds) given that the rope is at its lowest point when $t = 0$.
22. Write a trigonometric model for the average daily temperature in Birmingham, Alabama based on the graph to the right.

Source: National Climatic Data Center.



23. For the following equations:

- ☐ Graph one period of the function.
- ☐ Identify the characteristics (Domain, Range, X intercept(s), Y-intercept)
- ☐ Identify the characteristics (Amplitude, Midline, and Period)
- ☐ Identify any asymptotes, if necessary.

a. $f(x) = \cos(x)$

b. $g(x) = \sin(x)$

c. **Extension (not on test):** $h(x) = \tan(x)$

24. Evaluate and/or solve the following problems using the unit circle. *If unable to use the unit circle please use your graphing calculator.* Remember to check what mode you are in (radians vs. degrees).

a. $\sin(-\frac{5\pi}{6})$

i. $\sin(315^\circ)$

r. $\frac{1}{5} = \frac{1}{5}\cos \theta$ **

b. $\cos(\frac{7\pi}{6})$

j. $\sin(\frac{\pi}{12})$

s. $\tan \theta = -\sqrt{3}$

c. $\tan(0^\circ)$

k. $\tan(78^\circ)$

t. $\sec \theta = 1$ **

d. $\csc(\frac{11\pi}{4})$ **

l. $\sec(1.245)$

u. $1 = \tan \theta$

e. $\cos(180^\circ)$

m. $\sin \theta = \frac{1}{2}$

v. $\frac{4}{3} = 1 + \frac{1}{3}\tan \theta$ **

f. $\cos \frac{5\pi}{3}$

n. $-5 + 2 \tan \theta = -7$ **

w. $2 + \cos \theta = \frac{4 - \sqrt{2}}{2}$ **

g. $\sec(135^\circ)$ **

o. $\cos \theta = -1$

x. $\frac{\sqrt{2}}{2} = \sin \theta$

h. $\csc(\frac{5\pi}{4})$ **

p. $\sin \theta = 0$

q. $-\frac{\sqrt{3}}{2} = \sin \theta$